Week 8 — Activity: Archean Heat Production

Geophysics 3A: Potential Fields & Geothermics

May 5, 2026

Purpose

To connect radioactive decay, crustal differentiation, and geological processes to the observed heat production of ancient continental crust, we'll investigate the following question "Why is Archean heat production so low?"

Radiogenic decay

The concentration of a radioactive isotope is given by

$$C(t) = C_0 e^{-\lambda t},$$

and the corresponding heat production is

$$A(t) = \sum_i H_i \lambda_i C_i(t),$$

where H_i is the heat released per decay and λ_i is the decay constant.

Observation

Simple decay arguments predict higher heat production in the Archean than today, yet preserved Archean crust shows relatively modest values.

Tasks

For each mechanism below, explain qualitatively how it could reduce the heat production of preserved crustal rocks:

1. Differentiation and melt extraction of LILEs (K, Th, U).
2. Selective erosion of felsic upper crust.
3. Reworking: remelting and redistribution of existing crust.
4. Foundering of dense mafic lower crust or arc roots.
5. Secular evolution of mantle temperature.

Differentiation

Concept: Heat-producing elements (K, Th, U) behave as incompatible elements and strongly partition into melts.

Let $D \ll 1$ be the bulk partition coefficient of a heat-producing element. For fractional melting,

$$C_{\text{solid}} = C_0 (1 - F)^{(1/D - 1)},$$

where F is the melt fraction. You could think of this simply as:

$$C_{\text{crust}} \approx \frac{C_{\text{mantle}}}{D} F$$

Selective erosion

Concept: Heat production is concentrated in felsic upper crust, which is preferentially eroded.

Treat the crust as two reservoirs. Let A_u be upper crust heat production. If erosion removes heat-producing material at rate E,

$$\frac{dA_u}{dt} = -EA_u.$$

Solution:

$$A_u(t) = A_{u0}e^{-Et}$$

Reworking

Reworking here will be defined as reheating—past the closure temperatures of geochronometers—and remelting of existing crust rather than net addition of new crust.

Concept: Old crust is repeatedly melted, mixed, and redistributed during later tectonic events.

Schematic formulation: Model reworking as a mixing process. Let a fraction R of crust be reprocessed per event.

$$A_{n+1} = (1-R)A_n + RA_{\text{new}},$$

where A_{new} reflects heat production typical of newly generated crust.

Foundering

Concept: Dense, mafic, heat-poor material is preferentially removed from the lithosphere.

Let the lower crustal mass is reduced over time. If mafic lower crust is removed at rate F_d,

$$\frac{dM_{lc}}{dt} = -F_d M_{lc}.$$

The average crustal heat production becomes:

$$\bar{A}(t) = \frac{A_u M_u + A_{lc} M_{lc}(t)}{M_u + M_{lc}(t)}.$$

Secular cooling

Concept: The mantle was hotter early, affecting melt production rather than radiogenic inventory.

Mantle potential temperature declines approximately as:

$$T_m(t) = T_{m0}e^{-t/\tau},$$

where $\tau \sim$ 1–2 Ga is a thermal timescale. Melt production gives:

$$F(t) \propto T_m(t) - T_{\text{solidus}}.$$

Process	**Primary Physical Mechanism**	**Effect on Heat-Producing Elements (K, Th, U)**	**Primary Geological Consequence**
Radiogenic decay	Exponential decay of radioactive isotopes	Monotonic decrease of heat production with time	Higher heat production in the past; explains first-order time trend only
Differentiation and melt extraction	Incompatible elements partition strongly into melts ($D \ll 1$)	Rapid transfer of LILEs from mantle to crust early in Earth history	Early enrichment of crust; depletion of mantle
Selective erosion	Preferential removal of felsic, heat-rich upper crust	Progressive loss of the most heat-producing crustal components	Older crust appears cooler than expected from decay alone
Reworking (remelting and redistribution)	Repeated partial melting and crustal mixing	Dilution and homogenization of extreme heat production signatures	Preserved Archean crust is rarely pristine
Foundering of mafic lower crust	Removal of dense, heat-poor lower lithosphere	Bias toward preservation of felsic, buoyant crust	Thermally stable, thick continental lithosphere
Secular mantle thermal evolution	Early high mantle temperatures enhance melt production	Controls efficiency of LILE extraction rather than inventory	Strong early differentiation followed by declining crustal growth

Synthesis Questions

1. Which processes change the *distribution* of heat-producing elements?
2. Which processes affect what crust is *preserved*?
3. Why does radioactive decay alone fail to predict observed Archean heat production?

Which processes primarily:

- redistribute heat-producing elements?
- remove heat-producing material from the crust?
- bias what crust survives to the present day?

Reflection

Why is radioactive decay alone insufficient to explain Archean heat production patterns?

Concept Takeaway: Heat Production, Curvature, and Preservation

Radiogenic decay alone predicts that heat production should decrease smoothly with time, but observed Archean crust shows much lower heat production than this simple expectation.

The heat production of preserved continental crust reflects the combined effects of:

- *redistribution* of heat-producing elements through differentiation and melting,
- *removal* of heat-rich material by erosion and tectonic processes, and
- *preservation bias*, where buoyant, felsic, and thermally stable crust is more likely to survive.

In steady-state geotherms, these processes matter because crustal heat production controls the *curvature* of the geotherm:

$$T(z) = T_0 + \frac{q_0}{k} z - \frac{A}{2k} z^2.$$

Two regions with identical surface heat flow can therefore have very different Moho temperatures, lithospheric thicknesses, and long-term stability depending on how heat-producing elements were redistributed and preserved through time.